

DA Data Science and Artificial Intelligence

Section 1:	Probability and Statistics
Counting (permutation and combinations), probability axioms, Sample space, events, independent events, mutually exclusive events, marginal, conditional and joint probability, Bayes Theorem, conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, Continuous random variables and probability distribution function, uniform, exponential, Poisson, normal, standard normal, t-distribution, chi-squared distributions, cumulative distribution function, Conditional PDF, Central limit theorem, confidence interval, z-test, t-test, chi-squared test.	
Section 2:	Linear Algebra
Vector space, subspaces, linear dependence and independence of vectors, matrices, projection matrix, orthogonal matrix, idempotent matrix, partition matrix and their properties, quadratic forms, systems of linear equations and solutions; Gaussian elimination, eigenvalues and eigenvectors, determinant, rank, nullity, projections, LU decomposition, singular value decomposition.	
Section 3:	Calculus and Optimization
Functions of a single variable, limit, continuity and differentiability, Taylor series, maxima and minima, optimization involving a single variable.	
Section 4:	Programming, Data Structures and Algorithms
Programming in Python, basic data structures: stacks, queues, linked lists, trees, hash tables; Search algorithms: linear search and binary search, basic sorting algorithms: selection sort, bubble sort and insertion sort; divide and conquer: mergesort, quicksort; introduction to graph theory; basic graph algorithms: traversals and shortest path.	
Section 5:	Database Management and Warehousing
ER-model, relational model: relational algebra, tuple calculus, SQL, integrity constraints, normal form, file organization, indexing, data types, data transformation such as normalization, discretization, sampling, compression; data warehouse modelling: schema for multidimensional data models, concept hierarchies, measures: categorization and computations.	
Section 6:	Machine Learning
Supervised Learning: regression and classification problems, simple linear regression, multiple linear regression, ridge regression, logistic regression, k-nearest neighbour, naive Bayes classifier, linear discriminant analysis, support vector machine, decision trees, bias-variance trade-off, cross-validation methods such as leave-one-out (LOO) cross-validation, k-folds cross-validation, multi-layer perceptron, feed-forward neural network; Unsupervised Learning: clustering algorithms, k-means/k-medoid, hierarchical clustering, top-down, bottom-up: single-linkage, multiple-linkage, dimensionality reduction, principal component analysis.	
Section 7:	AI
Search: informed, uninformed, adversarial; logic, propositional, predicate; reasoning under uncertainty topics — conditional independence representation, exact inference through variable elimination, and approximate inference through sampling.	

General Aptitude

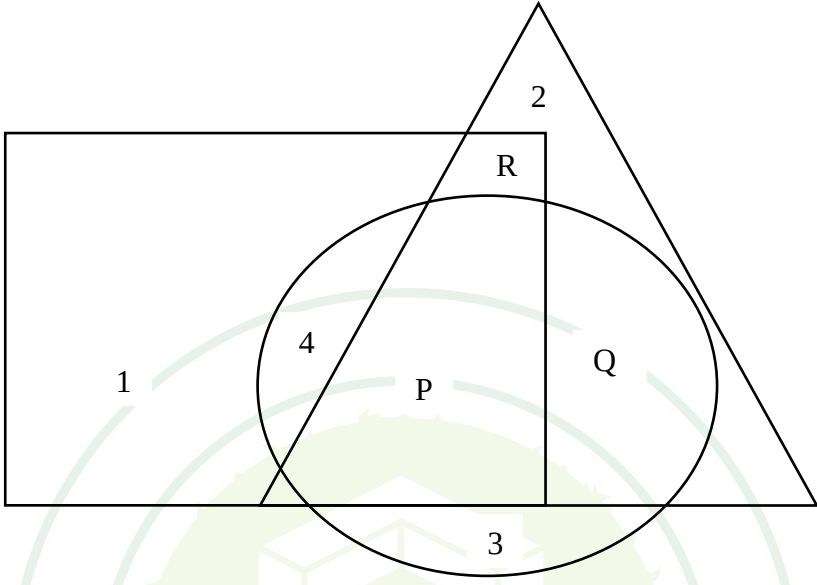
Q.1 – Q.5 Carry ONE mark Each

Q.1	Courage : Bravery :: Yearning : _____ Select the most appropriate option to complete the analogy.
(A)	Longing
(B)	Yelling
(C)	Yawning
(D)	Glaring

Q.2	We _____ tennis in the lawn when it suddenly started to rain. Select the most appropriate option to complete the above sentence.
(A)	have been playing
(B)	had been playing
(C)	would have been playing
(D)	could be playing

Q.3	A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:																
	<table><tr><td>0</td><td>1</td><td>0</td><td>2</td></tr><tr><td>4</td><td>7</td><td>3</td><td>3</td></tr><tr><td>5</td><td>5</td><td>4</td><td>4</td></tr><tr><td>6</td><td>7</td><td>3</td><td>2</td></tr></table>	0	1	0	2	4	7	3	3	5	5	4	4	6	7	3	2
0	1	0	2														
4	7	3	3														
5	5	4	4														
6	7	3	2														
(A)	3																
(B)	8																
(C)	11																
(D)	9																



Q.4	In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R ?
	
(A)	P = 6; Q = 5; R = 3
(B)	P = 5; Q = 6; R = 3
(C)	P = 3; Q = 6; R = 6
(D)	P = 5; Q = 3; R = 6

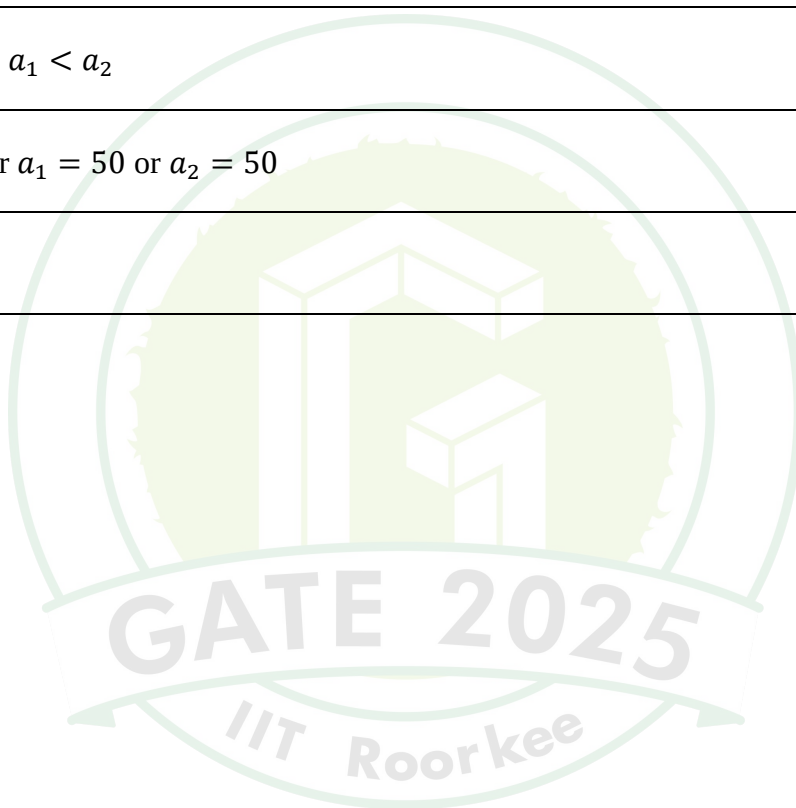
Q.5	A rectangle has a length L and a width W , where $L > W$. If the width, W , is increased by 10%, which one of the following statements is correct for all values of L and W ?
(A)	Perimeter increases by 10%.
(B)	Length of the diagonals increases by 10%.
(C)	Area increases by 10%.
(D)	The rectangle becomes a square.

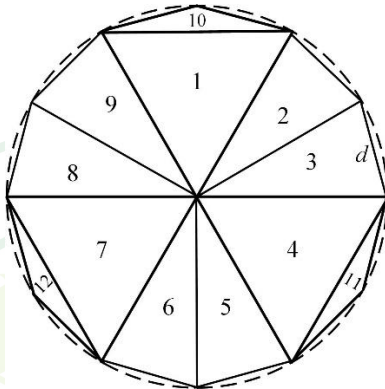


Q.6 – Q.10 Carry TWO marks Each

Q.6	Column-I has statements made by Shanthala; and, Column-II has responses given by Kanishk.			
	Column-I		Column-II	
	P.	This house is in a mess.	1.	Alright, I won't bring it up during our conversations.
	Q.	I am not happy with the marks given to me.	2.	Well, you can easily look it up.
	R.	Politics is a subject I avoid talking about.	3.	No problem, let me clear it up for you.
	S.	I don't know what this word means.	4.	Don't worry, I will take it up with your teacher.
Identify the option that has the correct match between Column-I and Column-II.				
(A)	P – 2; Q – 3; R – 1; S – 4			
(B)	P – 3; Q – 4; R – 1; S – 2			
(C)	P – 4; Q – 1; R – 2; S – 3			
(D)	P – 1; Q – 2; R – 4; S – 3			

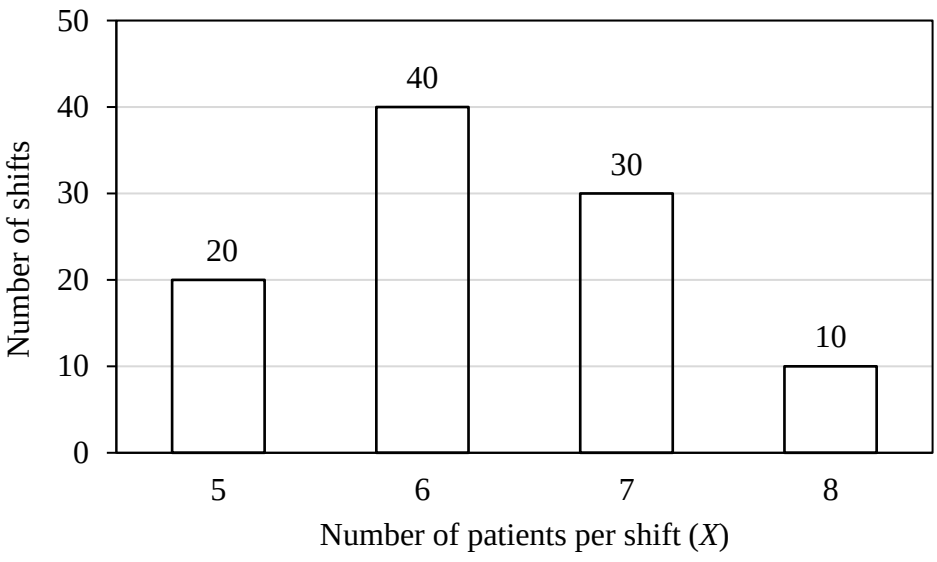
Q.7	Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight?
(A)	$a_1 < a_2 < 50$
(B)	$a_1 < 50 < a_2$
(C)	$50 < a_1 < a_2$
(D)	Either $a_1 = 50$ or $a_2 = 50$



Q.8	A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square. The number of squares that can be formed and the number of triangles required to form each square, respectively, are: Note: The figure shown is representative.
	
(A)	3; 4
(B)	4; 3
(C)	3; 3
(D)	3; 2

Q.9	If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:
(A)	2^{-1}
(B)	2^0
(C)	2^3
(D)	2^{15}



Q.10	The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is ₹ $1000(X - 0.2)$, what is the average amount (in ₹) she has earned per shift in the past 100 shifts? Note: The figure shown is representative.										
	 <table border="1"> <thead> <tr> <th>Number of patients per shift (X)</th> <th>Number of shifts</th> </tr> </thead> <tbody> <tr> <td>5</td> <td>20</td> </tr> <tr> <td>6</td> <td>40</td> </tr> <tr> <td>7</td> <td>30</td> </tr> <tr> <td>8</td> <td>10</td> </tr> </tbody> </table>	Number of patients per shift (X)	Number of shifts	5	20	6	40	7	30	8	10
Number of patients per shift (X)	Number of shifts										
5	20										
6	40										
7	30										
8	10										
(A)	6,100										
(B)	6,300										
(C)	6,000										
(D)	6,500										

Q. 11 – Q. 35 carry one mark each.

- Q. 11 Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X|Y]$. Then $E[E[X|Y]]$ equals
- (A) $E[X|Y]$
 - (B) $\frac{E[X]}{E[Y]}$
 - (C) $E[X]$
 - (D) $E[Y]$
- Q. 12 The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order
- (A) $O(n)$
 - (B) $O(n^2)$
 - (C) $O(n^3)$
 - (D) $O(n^4)$
- Q. 13 The sum of the elements in each row of $A \in \mathbb{R}^{n \times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?
- (A) The equation $Bx = 0$ has no solution
 - (B) The equation $Bx = 0$ has exactly two solutions
 - (C) The equation $Bx = 0$ has infinitely many solutions
 - (D) The equation $Bx = 0$ has a unique solution

- Q. 14 Let $f(x) = \frac{e^x - e^{-x}}{2}$, $x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a . What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial)
- (A) 0
- (B) 1
- (C) $\frac{1}{10!}$
- (D) $\frac{2}{10!}$
- Q. 15 Let p and q be any two propositions. Consider the following propositional statements.
 $S_1 : p \rightarrow q$, $S_2 : \neg p \wedge q$, $S_3 : \neg p \vee q$, $S_4 : \neg p \vee \neg q$,
where $\wedge$ denotes conjunction (AND operation), $\vee$ denotes disjunction (OR operation),
and $\neg$ denotes negation (NOT operation). Which one of the following options is correct?
(Note: $\equiv$ denotes logical equivalence)
- (A) $S_1 \equiv S_3$
- (B) $S_2 \equiv S_3$
- (C) $S_2 \equiv S_4$
- (D) $S_1 \equiv S_4$
- Q. 16 If a relational decomposition is not dependency-preserving, which one of the following relational operators will be executed more frequently in order to maintain the dependencies?
- (A) Selection
- (B) Projection
- (C) Join
- (D) Set union

Q. 17 Consider the following three relations:

Car (model, year, serial, color)

Make (maker, model)

Own (owner, serial)

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}}(\text{Own} \bowtie (\sigma_{\text{color}=\text{"red"}}(\text{Car} \bowtie (\sigma_{\text{maker}=\text{"ABC"}}\text{Make}))))$$

Which one of the following options describes what the above expression computes?

- (A) All owners of a red car, a car made by ABC, or a red car made by ABC
- (B) All owners of more than one car, where at least one car is red and made by ABC
- (C) All owners of a red car made by ABC
- (D) All red cars made by ABC

Q. 18 Consider a hash table of size 10 with indices $\{0, 1, \dots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10},$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively

- (A) 2 and 5
- (B) 2 and 6
- (C) 4 and 5
- (D) 4 and 6

- Q. 19 Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

If the median of X is 3, then what is the value of t ?

- (A) 2
- (B) 1
- (C) -1
- (D) 0
- Q. 20 Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4,$$

where $E[X]$ denotes the expectation of random variable X . The values of a, b are:

- (A) $a = -2, b = 1$
- (B) $a = 2, b = -1$
- (C) $a = -2, b = -1$
- (D) $a = 1, b = 1$

- Q. 21 It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X . What is the value of λ ?
($\ln$ denotes natural logarithm)

- (A) $\ln 2$
(B) $\ln 4$
(C) $\ln 3$
(D) $\ln 0.25$

- Q. 22 Consider designing a linear classifier

$$y = \text{sign}(f(x; w, b)), \quad f(x; w, b) = w^\top x + b$$

on a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$, $x_i \in \mathbb{R}^d$, $y_i \in \{+1, -1\}$, $i = 1, 2, \dots, N$. Recall that the sign function outputs +1 if the argument is positive, and -1 if the argument is non-positive. The parameters w and b are updated as per the following training algorithm:

$$w_{\text{new}} = w_{\text{old}} + y_n x_n, \quad b_{\text{new}} = b_{\text{old}} + y_n$$

whenever $\text{sign}(f(x_n; w_{\text{old}}, b_{\text{old}})) \neq y_n$. In other words, whenever the classifier wrongly predicts a sample (x_n, y_n) from the dataset, w_{old} gets updated to w_{new} , and likewise b_{old} gets updated to b_{new} . Consider the case $(x_n, +1)$, $f(x_n; w_{\text{old}}, b_{\text{old}}) < 0$. Then

- (A) $f(x_n; w_{\text{new}}, b_{\text{new}}) > f(x_n; w_{\text{old}}, b_{\text{old}})$
(B) $f(x_n; w_{\text{new}}, b_{\text{new}}) < f(x_n; w_{\text{old}}, b_{\text{old}})$
(C) $f(x_n; w_{\text{new}}, b_{\text{new}}) = f(x_n; w_{\text{old}}, b_{\text{old}})$
(D) $y_n f(x_n; w_{\text{old}}, b_{\text{old}}) > 1$

Q. 23 Consider the following Python declarations of two lists.

$A = [1, 2, 3]$

$B = [4, 5, 6]$

Which one of the following statements results in $A = [1, 2, 3, 4, 5, 6]$?

(A) $A.extend(B)$

(B) $A.append(B)$

(C) $A.update(B)$

(D) $A.insert(B)$

Q. 24 Consider two functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow (1, \infty)$. Both functions are differentiable at a point c . Which of the following functions is/are ALWAYS differentiable at c ? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions.

(A) $f \pm g$

(B) $f \cdot g$

(C) $\frac{f}{g}$

(D) $f \circ g + g \circ f$

Q. 25 Which of the following statements is/are correct?

(A) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors

(B) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

(C) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

(D) Orthonormal vectors $\mathbb{R}^n$ are linearly independent

- Q. 26 Which of the following statements is/are correct in a Bayesian network?
- (A) Variable elimination is an approximate inference algorithm
 - (B) Gibbs sampling is an exact inference algorithm
 - (C) Variable elimination is used to determine conditional probabilities
 - (D) Rejection sampling is an approximate inference algorithm
- Q. 27 For which of the following inputs does binary search take time $O(\log n)$ in the worst case?
- (A) An array of n integers in any order
 - (B) A linked list of n integers in any order
 - (C) An array of n integers in increasing order
 - (D) A linked list of n integers in increasing order
- Q. 28 Let $A = I_n + xx^\top$, where I_n is the $n \times n$ identity matrix and $x \in \mathbb{R}^n$, $x^\top x = 1$. Which of the following options is/are correct?
- (A) Rank of A is n
 - (B) A is invertible
 - (C) 0 is an eigenvalue of A
 - (D) A^{-1} has a negative eigenvalue
- Q. 29 Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x .
- (A) 10
 - (B) 12
 - (C) 14
 - (D) 16

- Q. 30 Let C_1 and C_2 be two sets of objects. Let $D(x, y)$ be a measure of dissimilarity between two objects x and y . Consider the following definitions of dissimilarity between C_1 and C_2 .

$$\mathbf{DIS-1}(C_1, C_2) = \max_{x \in C_1, y \in C_2} D(x, y)$$

$$\mathbf{DIS-2}(C_1, C_2) = \min_{x \in C_1, y \in C_2} D(x, y)$$

Which of the following statements is/are correct?

- (A) Single Linkage Clustering uses **DIS-1**
 - (B) Single Linkage Clustering uses **DIS-2**
 - (C) Complete Linkage Clustering uses **DIS-2**
 - (D) Complete Linkage Clustering uses **DIS-1**
- Q. 31 There are three boxes containing white balls and black balls.

Box-1 contains 2 black and 1 white balls.

Box-2 contains 1 black and 2 white balls.

Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____ (Round off to two decimal places)

- Q. 32 $\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t =$ _____
(Round off to one decimal place)

Q. 33 On a relation named **Loan** of a bank:

Loan		
loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

the following SQL query is executed.

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
                   FROM Loan L2
                   WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____ (Answer in integer)

Q. 34 Given data $\{(-1, 1), (2, -5), (3, 5)\}$ of the form (x, y) , we fit a model $y = wx$ using linear least-squares regression. The optimal value of w is _____
(Round off to three decimal places)

Q. 35 The naive Bayes classifier is used to solve a two-class classification problem with class-labels y_1, y_2 . Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with

$$P(x|y_1) = \frac{3}{4} \quad \text{and} \quad P(x|y_2) = \frac{1}{4}$$

for a specific feature vector x . The probability of misclassifying x is _____
(Round off to two decimal places)

Q. 36 – Q. 65 carry two marks each.

- Q. 36 Let $Y = Z^2$, $Z = \frac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2 . The variance of Y is
- (A) 1
- (B) 2
- (C) 3
- (D) 4
- Q. 37 Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?
- (A) A is invertible
- (B) Determinant of A is 0
- (C) The sum of the diagonal elements of A is 1
- (D) A and A^2 have the same rank
- Q. 38 Let $\{x_1, x_2, \dots, x_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j) -th element of matrix $A \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = x_i^\top x_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct?
- (A) A is invertible
- (B) 0 is a singular value of A
- (C) Determinant of A is 0
- (D) $z^\top A z = 0$ for some non-zero $z \in \mathbb{R}^n$

Q. 39 Consider the cumulative distribution function (CDF) of a random variable X :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases}$$

The value of $P(X^2 \leq 0.25)$ is

- (A) 0.625
- (B) 0.25
- (C) 0.5
- (D) 0.5625

Q. 40 A random variable X is said to be distributed as $Bernoulli(\theta)$, denoted by $X \sim Bernoulli(\theta)$, if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim Bernoulli(\theta)$, $i = 1, 2, \dots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through Central Limit Theorem, is given by

(Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$)

- (A) $\phi(2) - \phi(-2)$
- (B) $\phi(1) - \phi(-1)$
- (C) $\phi(3) - \phi(-3)$
- (D) $\phi(90) - \phi(60)$

Q. 41 For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

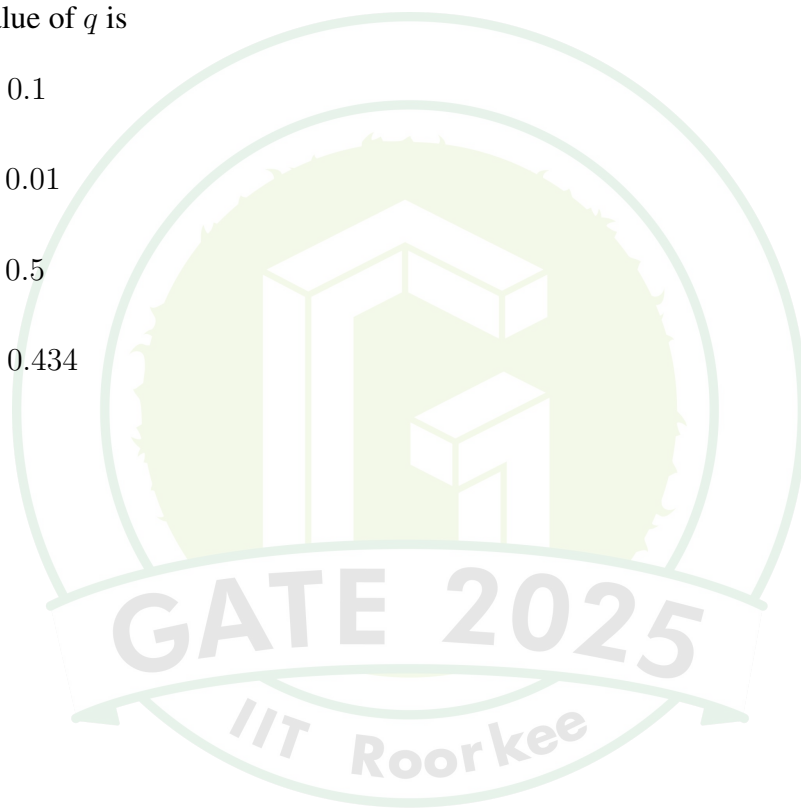
$$\lfloor x \rfloor = k, \quad k \leq x < k + 1,$$

where k is an integer. Let $Y = \lfloor X \rfloor$, where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where $\ln$ denotes natural logarithm. For any positive integer ℓ , one can write the probability of the event $Y = \ell$ as follows

$$P(Y = \ell) = q^\ell(1 - q)$$

The value of q is

- (A) 0.1
- (B) 0.01
- (C) 0.5
- (D) 0.434



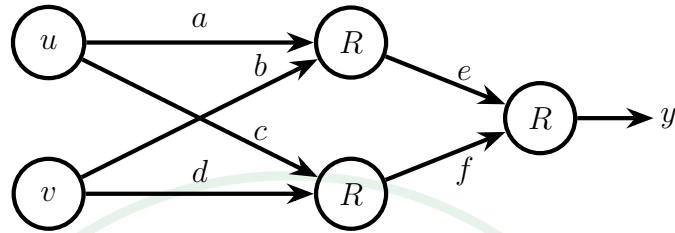
Q. 42 Consider the neural network shown in the figure with

inputs: u, v

weights: a, b, c, d, e, f

output: y

R denotes the ReLU function, $R(x) = \max(0, x)$.



Given $u = 2, v = 3$,

$a = 1, b = 1, c = 1, d = -1, e = 4, f = -1$,

which one of the following is correct?

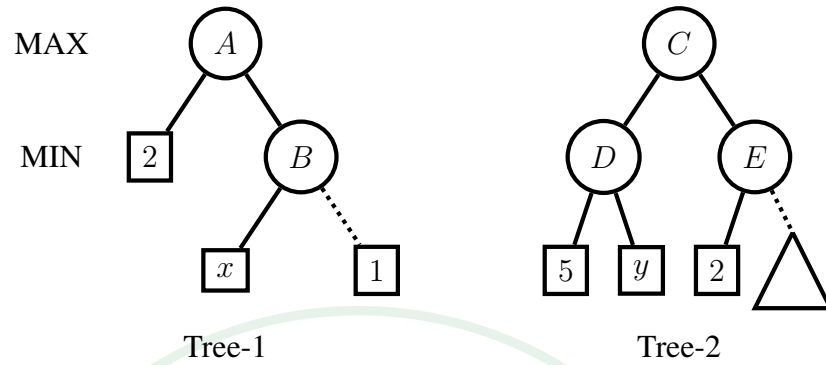
(A) $\frac{\partial y}{\partial a} = 8, \frac{\partial y}{\partial f} = 0$

(B) $\frac{\partial y}{\partial a} = 1, \frac{\partial y}{\partial f} = 0$

(C) $\frac{\partial y}{\partial a} = 1, \frac{\partial y}{\partial f} = -1$

(D) $\frac{\partial y}{\partial a} = 2, \frac{\partial y}{\partial f} = -1$

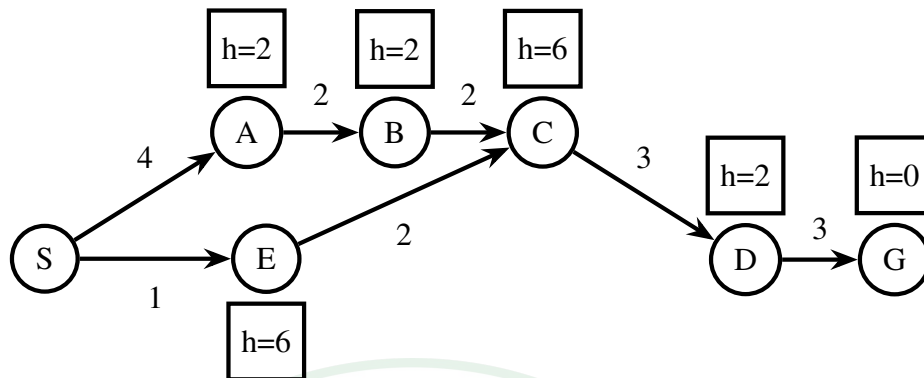
- Q. 43 Consider game trees Tree-1 and Tree-2 as shown. The first level is a MAX agent and the second level is a MIN agent. The value in the square node is the output of the utility function.



For what ranges of x and y , the right child of node B and the right child of node E will be pruned by alpha-beta pruning algorithm?

- (A) $x \in [1, \infty)$ and $y \in (-\infty, 2]$
- (B) $x \in (-\infty, 2]$ and $y \in (-\infty, 5]$
- (C) $x \in (-\infty, 2]$ and $y \in [2, \infty)$
- (D) $x \in [1, \infty)$ and $y \in (-\infty, 5]$

- Q. 44 The state graph shows the action cost along the edges and the heuristic function h associated with each state.



Suppose A^* algorithm is applied on this state graph using priority queue to store the frontier. In what sequence are the nodes expanded?

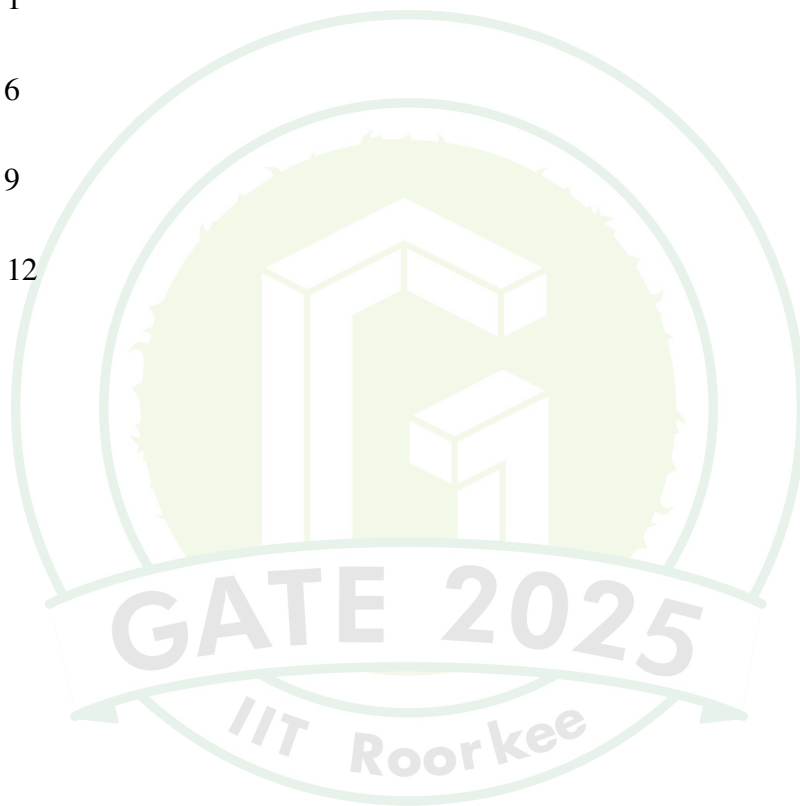
- (A) S,A,E,C,B,D,G
 (B) S,E,A,C,B,D,G
 (C) S,A,E,B,C,D,G
 (D) S,A,B,E,C,D,G
- Q. 45 A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

- (A) 0
 (B) 1
 (C) $1 - \left(\frac{5}{6}\right)^{100}$
 (D) $\left(\frac{5}{6}\right)^{100}$

- Q. 46 Consider a fact table in an OLAP application: $\text{Facts}(D1, D2, \text{val})$, where $D1$ and $D2$ are its dimension attributes and val is a dependent attribute. Suppose attribute $D1$ takes 3 values and $D2$ takes 2 values, and all combinations of these values are present in the table Facts . How many tuples are there in the result of the following query?

```
SELECT D1, D2, sum(val)
FROM Facts
GROUP BY CUBE (D1, D2);
```

- (A) 1
- (B) 6
- (C) 9
- (D) 12



Q. 47 Consider the following Python code snippet.

```
A={"this", "that"}
B={"that", "other"}
C={"other", "this"}
while "other" in C:
    if "this" in A:
        A, B, C=A-B, B-C, C-A
    if "that" in B:
        A, B, C=C|A, A|B, B|C
```

When the above program is executed, at the end, which of the following sets contains "this"?

- (A) Only A
- (B) Only B
- (C) Only C
- (D) A, C

Q. 48 Which of the following statements is/are correct about the rectified linear unit (ReLU) activation function defined as $\text{ReLU}(x) = \max(x, 0)$, where $x \in \mathbb{R}$?

- (A) ReLU is continuous everywhere
- (B) ReLU is differentiable everywhere
- (C) ReLU is not differentiable at $x = 0$
- (D) $\text{ReLU}(x) = \text{ReLU}(ax)$, for all $a \in \mathbb{R}$

- Q. 49 Consider the function $f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}$, $x \in [-8, 0]$. Which of the following statements is/are correct?
- (A) The maximum value of f is attained at $x = -5$
 - (B) The minimum value of f is attained at $x = -2$
 - (C) The maximum value of f is $\frac{133}{2}$
 - (D) The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$
- Q. 50 Let x_1, x_2, x_3, x_4, x_5 be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $A = x_1x_1^\top + \dots + x_5x_5^\top$. Which of the following statements is/are correct?
- (A) Singular values of A are also its eigenvalues
 - (B) Singular values of A are either 0 or 1
 - (C) Determinant of A is 1
 - (D) A is invertible
- Q. 51 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct?
- (A) f has a local minima
 - (B) There does not exist x and y , $x \neq y$, such that $f'(x) = f'(y) = 0$
 - (C) f has at most one global minimum
 - (D) f has at most one local minimum

Q. 52 An $n \times n$ matrix A with real entries satisfies the property: $\|Ax\|^2 = \|x\|^2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct?

- (A) A must be orthogonal
- (B) $A = I$, where I denotes the identity matrix, is the only solution
- (C) The eigenvalues of A are either +1 or -1
- (D) A has full rank

Q. 53 Consider designing a linear binary classifier $f(x) = \text{sign}(w^\top x + b)$, $x \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1: } \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \text{Class-2: } \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

Hard-margin support vector machine (SVM) formulation is solved to obtain w and b . Which of the following options is/are correct?

- (A) $w = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = 1$
- (B) The number of support vectors is 3
- (C) The margin is $\sqrt{2}$
- (D) Training accuracy is 98%

- Q. 54 Consider a coin-toss experiment where the probability of head showing up is p . In the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

where n is the total number of independent coin tosses.

Which of the following statements is/are correct?

- (A) $E[\hat{p}] = p$
- (B) $E[\hat{p}] = \frac{p}{n}$
- (C) As n increases, variance of $\hat{p}$ decreases
- (D) Variance of $\hat{p}$ does not depend on n

- Q. 55 Consider a two-class problem in $\mathbb{R}^d$ with class labels *red* and *green*. Let μ_{red} and μ_{green} be the means of the two classes. Given test sample $x \in \mathbb{R}^d$, a classifier calculates the squared Euclidean distance (denoted by $\|\cdot\|^2$) between x and the means of the two classes and assigns the class label that the sample x is closest to. That is, the classifier computes

$$f(x) = \|\mu_{red} - x\|^2 - \|\mu_{green} - x\|^2$$

and assigns the label *red* to x if $f(x) < 0$, and *green* otherwise. Which of the following statements is/are correct?

- (A) The sample $x = 0$ is assigned the label *green* if $\|\mu_{red}\| < \|\mu_{green}\|$
- (B) f is a linear function of x
- (C) $f(x) = w^\top x + b$, where w and b are functions of μ_{red} and μ_{green}
- (D) f is a quadratic polynomial in x

Q. 56 Consider the following two relations, named `Customer` and `Person`, in a database:

`Person` (

`aadhaar CHAR(12) PRIMARY KEY,`

`name VARCHAR(32));`

`Customer` (

`name VARCHAR(32),`

`email VARCHAR(32) PRIMARY KEY,`

`phone CHAR(10),`

`aadhaar CHAR(12),`

`FOREIGN KEY (aadhaar) REFERENCES Person(aadhaar));`

Which of the following statements is/are correct?

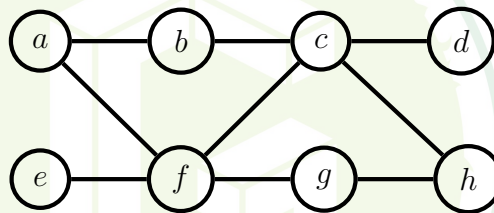
- (A) `aadhaar` is a candidate key in the `Customer` relation
- (B) `phone` can be NULL in the `Customer` relation
- (C) `aadhaar` is a candidate key in the `Person` relation
- (D) `aadhaar` can be NULL in the `Person` relation

- Q. 57 Consider a database relation R with attributes $ABCDEFG$, and having the following functional dependencies:

$$A \rightarrow BCEF \quad E \rightarrow DG \quad BC \rightarrow A$$

Which of the following statements is/are correct?

- (A) A is the only candidate key of R
- (B) A, BC are the candidate keys of R
- (C) A, BC, E are the candidate keys of R
- (D) Relation R is not in Boyce-Codd Normal Form (BCNF)
- Q. 58 Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.



It is given that $a - b - c - d$ is a shortest path between a and d ; $e - f - g - h$ is a shortest path between e and h ; $a - f - c - h$ is a shortest path between a and h . Which of the following is/are NOT the edges of G ?

- (A) (b, d)
- (B) (b, g)
- (C) (b, h)
- (D) (e, g)
- Q. 59 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then $f(1) - f(0) = \underline{\hspace{2cm}}$ (Answer in integer)

- Q. 60 Let $D = \{x^{(1)}, \dots, x^{(n)}\}$ be a dataset of n observations where each $x^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^n x^{(i)} = 0$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \leq i \leq 100$. Let $u \in \mathbb{R}^{100}$ be the direction of maximum variance with $u^\top u = 1$.

The value of

$$\frac{1}{n} \sum_{i=1}^n (u^\top x^{(i)})^2 = \underline{\hspace{2cm}}$$

(Answer in integer)

- Q. 61 A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment.

The expectation of S_{100} denoted by $E[S_{100}] = \underline{\hspace{2cm}}$

(Round off to one decimal place)

- Q. 62 Consider the following tables, **Loan** and **Borrower**, of a bank.

Loan		
loan_num	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower	
customer_name	loan_num
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Query: $\pi_{\text{branch_name}, \text{customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan})$

where $\bowtie$ denotes natural join.

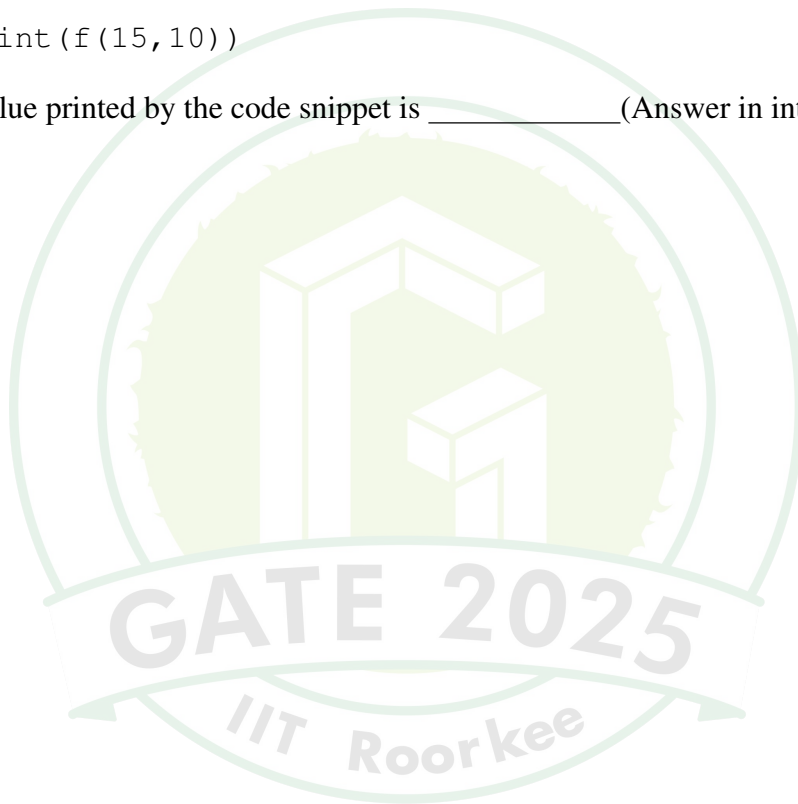
The number of tuples returned by the above relational algebra query is $\underline{\hspace{2cm}}$

(Answer in integer)

Q. 63 Consider the following Python code snippet.

```
def f(a,b):  
    if (a==0):  
        return b  
    if (a%2==1):  
        return 2*f((a-1)/2,b)  
    return b+f(a-1,b)  
  
print(f(15,10))
```

The value printed by the code snippet is _____ (Answer in integer)



Q. 64 Consider the following pseudocode.

```
Create empty stack S
Set x=0, flag=0, sum=0
Push x onto S
while (S is not empty){
    if (flag equals 0){
        Set x = x+1
        Push x onto S
    }
    if (x equals 8):
        Set flag=1
    if (flag equals 1){
        x = Pop(S)
        if (x is odd):
            Pop(S)
        Set sum = sum + x
    }
}
Output sum
```

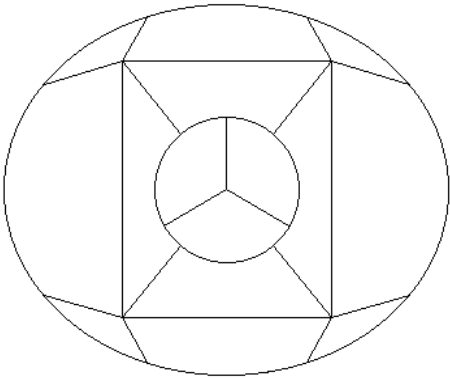
The value of `sum` output by a program executing the above pseudocode is _____
(Answer in integer)

Q. 65 Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \dots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2, \text{ for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in *decreasing* order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____ (Answer in integer)

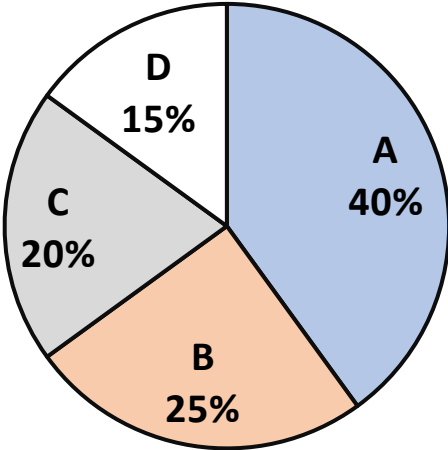
General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1	If ‘→’ denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft]. Which one of the given options is appropriate to fill the blank?
(A)	frown
(B)	fawn
(C)	vein
(D)	vain

Q.2	The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is 
(A)	4
(B)	3
(C)	5
(D)	6

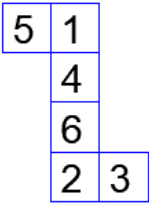
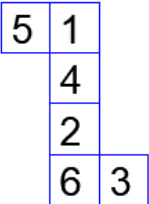
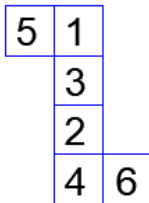
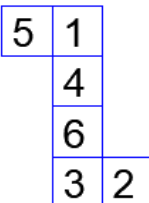
Q.3	How many 4-digit positive integers divisible by 3 can be formed using only the digits {1, 3, 4, 6, 7}, such that no digit appears more than once in a number?
(A)	24
(B)	48
(C)	72
(D)	12
Q.4	The sum of the following infinite series is $2 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{8} + \frac{1}{9} + \frac{1}{16} + \frac{1}{27} + \dots$
(A)	11/3
(B)	7/2
(C)	13/4
(D)	9/2

Q.5	In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid. Share of valid votes  Based on the data provided, the total number of valid votes received by the candidates B and C is
(A)	45,000
(B)	49,500
(C)	51,750
(D)	54,000

Q.6 – Q.10 Carry TWO marks Each

Q.6	Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?
(A)	All human beings can digest dairy milk.
(B)	No human being can digest dairy milk.
(C)	Digestion of dairy milk is essential for human beings.
(D)	In human beings, digestion of dairy milk resulted from a mutated gene.
Q.7	The probability of a boy or a girl being born is $\frac{1}{2}$. For a family having only three children, what is the probability of having two girls and one boy?
(A)	$\frac{3}{8}$
(B)	$\frac{1}{8}$
(C)	$\frac{1}{4}$
(D)	$\frac{1}{2}$

Q.8	Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table. <table><tr><td></td><td>Mutual fund A</td><td>Mutual fund B</td><td>Mutual fund C</td></tr><tr><td>Person 1</td><td>₹10,000</td><td>₹20,000</td><td>₹20,000</td></tr><tr><td>Person 2</td><td>₹20,000</td><td>₹15,000</td><td>₹15,000</td></tr></table> At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?		Mutual fund A	Mutual fund B	Mutual fund C	Person 1	₹10,000	₹20,000	₹20,000	Person 2	₹20,000	₹15,000	₹15,000
	Mutual fund A	Mutual fund B	Mutual fund C										
Person 1	₹10,000	₹20,000	₹20,000										
Person 2	₹20,000	₹15,000	₹15,000										
(A)	7.5%												
(B)	10%												
(C)	15%												
(D)	20%												

Q.9	Three different views of a dice are shown in the figure below.    The piece of paper that can be folded to make this dice is
(A)	
(B)	
(C)	
(D)	

Q.10	Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?
(A)	A rhombus
(B)	A triangle
(C)	An ellipse
(D)	A hexagon

Q.11 – Q.35 Carry ONE mark Each

Q.11	Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which ONE of the following options is correct?
(A)	Both (i) and (ii) are true
(B)	(i) is true and (ii) is false
(C)	(ii) is true and (i) is false
(D)	Both (i) and (ii) are false

Q.12	Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails. What is the probability of the event $T \cap S$?
(A)	0
(B)	0.5
(C)	0.25
(D)	1

Q.13	Consider the matrix $M = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$. Which ONE of the following statements is TRUE ?
(A)	The eigenvalues of M are non-negative and real.
(B)	The eigenvalues of M are complex conjugate pairs.
(C)	One eigenvalue of M is positive and real, and another eigenvalue of M is zero.
(D)	One eigenvalue of M is non-negative and real, and another eigenvalue of M is negative and real.

Q.14	Consider performing depth-first search (DFS) on an undirected and unweighted graph G starting at vertex s . For any vertex u in G , $d[u]$ is the length of the shortest path from s to u . Let (u, v) be an edge in G such that $d[u] < d[v]$. If the edge (u, v) is explored first in the direction from u to v during the above DFS, then (u, v) becomes a _____ edge.
(A)	tree
(B)	cross
(C)	back
(D)	gray

Q.15	For any twice differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$, if at some $x^* \in \mathbb{R}$, $f'(x^*) = 0$ and $f''(x^*) > 0$, then the function f necessarily has a _____ at $x = x^*$. Note: $\mathbb{R}$ denotes the set of real numbers.
(A)	local minimum
(B)	global minimum
(C)	local maximum
(D)	global maximum

Q.16	Match the items in Column 1 with the items in Column 2 in the following table:								
	<table border="1"> <thead> <tr> <th>Column 1</th><th>Column 2</th></tr> </thead> <tbody> <tr> <td>(p) First In First Out</td><td>(i) Stacks</td></tr> <tr> <td>(q) Lookup Operation</td><td>(ii) Queues</td></tr> <tr> <td>(r) Last In First Out</td><td>(iii) Hash Tables</td></tr> </tbody> </table>	Column 1	Column 2	(p) First In First Out	(i) Stacks	(q) Lookup Operation	(ii) Queues	(r) Last In First Out	(iii) Hash Tables
Column 1	Column 2								
(p) First In First Out	(i) Stacks								
(q) Lookup Operation	(ii) Queues								
(r) Last In First Out	(iii) Hash Tables								
(A)	(p) – (ii), (q) – (iii), (r) – (i)								
(B)	(p) – (ii), (q) – (i), (r) – (iii)								
(C)	(p) – (i), (q) – (ii), (r) – (iii)								
(D)	(p) – (i), (q) – (iii), (r) – (ii)								

Q.17	Consider the dataset with six datapoints: $\{(x_1, y_1), (x_2, y_2), \dots, (x_6, y_6)\}$, where $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $x_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $x_5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $x_6 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ and the labels are given by $y_1 = y_2 = y_5 = 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear support vector machine is trained on the above dataset. Which ONE of the following sets is a possible set of support vectors?
(A)	$\{x_1, x_2, x_5\}$
(B)	$\{x_3, x_4, x_5\}$
(C)	$\{x_4, x_5\}$
(D)	$\{x_1, x_2, x_3, x_4\}$

Q.18	Match the items in Column 1 with the items in Column 2 in the following table: <table border="1"> <thead> <tr> <th>Column 1</th><th>Column 2</th></tr> </thead> <tbody> <tr> <td>(p) Principal Component Analysis</td><td>(i) Discriminative Model</td></tr> <tr> <td>(q) Naïve Bayes Classification</td><td>(ii) Dimensionality Reduction</td></tr> <tr> <td>(r) Logistic Regression</td><td>(iii) Generative Model</td></tr> </tbody> </table>	Column 1	Column 2	(p) Principal Component Analysis	(i) Discriminative Model	(q) Naïve Bayes Classification	(ii) Dimensionality Reduction	(r) Logistic Regression	(iii) Generative Model
Column 1	Column 2								
(p) Principal Component Analysis	(i) Discriminative Model								
(q) Naïve Bayes Classification	(ii) Dimensionality Reduction								
(r) Logistic Regression	(iii) Generative Model								
(A)	(p) – (iii), (q) – (i), (r) – (ii)								
(B)	(p) – (ii), (q) – (i), (r) – (iii)								
(C)	(p) – (ii), (q) – (iii), (r) – (i)								
(D)	(p) – (iii), (q) – (ii), (r) – (i)								

Q.19	Euclidean distance based k -means clustering algorithm was run on a dataset of 100 points with $k = 3$. If the points $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ are both part of cluster 3, then which ONE of the following points is necessarily also part of cluster 3?
(A)	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
(B)	$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
(C)	$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
(D)	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Q.20	Given a dataset with K binary-valued attributes (where $K > 2$) for a two-class classification task, the number of parameters to be estimated for learning a naïve Bayes classifier is
(A)	$2^K + 1$
(B)	$2K + 1$
(C)	$2^{K+1} + 1$
(D)	$K^2 + 1$

Q.21	Consider performing uniform hashing on an open address hash table with load factor $\alpha = \frac{n}{m} < 1$, where n elements are stored in the table with m slots. The expected number of probes in an unsuccessful search is at most $\frac{1}{1-\alpha}$. Inserting an element in this hash table requires at most _____ probes, on average.
(A)	$\ln\left(\frac{1}{1-\alpha}\right)$
(B)	$\frac{1}{1-\alpha}$
(C)	$1 + \frac{\alpha}{2}$
(D)	$\frac{1}{1+\alpha}$

Q.22	For any binary classification dataset, let $S_B \in \mathbb{R}^{d \times d}$ and $S_W \in \mathbb{R}^{d \times d}$ be the between-class and within-class scatter (covariance) matrices, respectively. The Fisher linear discriminant is defined by $u^* \in \mathbb{R}^d$, that maximizes $J(u) = \frac{u^T S_B u}{u^T S_W u}$ If $\lambda = J(u^*)$, S_W is non-singular and $S_B \neq 0$, then (u^*, λ) must satisfy which ONE of the following equations? Note: $\mathbb{R}$ denotes the set of real numbers.
(A)	$S_W^{-1} S_B u^* = \lambda u^*$
(B)	$S_W u^* = \lambda S_B u^*$
(C)	$S_B S_W u^* = \lambda u^*$
(D)	$u^{*T} u^* = \lambda^2$

Q.23	Let h_1 and h_2 be two admissible heuristics used in A^* search. Which ONE of the following expressions is always an admissible heuristic?
(A)	$h_1 + h_2$
(B)	$h_1 \times h_2$
(C)	$h_1/h_2, (h_2 \neq 0)$
(D)	$ h_1 - h_2 $

Q.24	Consider five random variables U, V, W, X , and Y whose joint distribution satisfies: $P(U, V, W, X, Y) = P(U)P(V)P(W U, V)P(X W)P(Y W)$ Which ONE of the following statements is FALSE ?
(A)	Y is conditionally independent of V given W
(B)	X is conditionally independent of U given W
(C)	U and V are conditionally independent given W
(D)	Y and X are conditionally independent given W

Q.25	Consider the following statement: In adversarial search, α-β pruning can be applied to game trees of any depth where α is the <u>(m)</u> value choice we have formed so far at any choice point along the path for the MAX player and β is the <u>(n)</u> value choice we have formed so far at any choice point along the path for the MIN player. Which ONE of the following choices of (m) and (n) makes the above statement valid?
(A)	(m) = highest, (n) = highest
(B)	(m) = lowest, (n) = highest
(C)	(m) = highest, (n) = lowest
(D)	(m) = lowest, (n) = lowest

Q.26	Consider a database that includes the following relations: Defender(<i>name, rating, side, goals</i>) Forward(<i>name, rating, assists, goals</i>) Team(<i>name, club, price</i>) Which ONE of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where ϕ denotes the empty set?
(A)	$\Pi_{name}(Team) \setminus (\Pi_{name}(Defender) \cap \Pi_{name}(Forward)) = \phi$
(B)	$(\Pi_{name}(Defender) \cap \Pi_{name}(Forward)) \setminus \Pi_{name}(Team) = \phi$
(C)	$\Pi_{name}(Team) \setminus (\Pi_{name}(Defender) \cup \Pi_{name}(Forward)) = \phi$
(D)	$(\Pi_{name}(Defender) \cup \Pi_{name}(Forward)) \setminus \Pi_{name}(Team) = \phi$

Q.27	Let the minimum, maximum, mean and standard deviation values for the attribute <i>income</i> of data scientists be ₹46000, ₹170000, ₹96000, and ₹21000, respectively. The z-score normalized <i>income</i> value of ₹106000 is closest to which ONE of the following options?
(A)	0.217
(B)	0.476
(C)	0.623
(D)	2.304

Q.28	Consider the following tree traversals on a full binary tree: (i) Preorder (ii) Inorder (iii) Postorder Which of the following traversal options is/are sufficient to uniquely reconstruct the full binary tree?
(A)	(i) and (ii)
(B)	(ii) and (iii)
(C)	(i) and (iii)
(D)	(ii) only

Q.29	Let x and y be two propositions. Which of the following statements is a tautology /are tautologies?
(A)	$(\neg x \wedge y) \Rightarrow (y \Rightarrow x)$
(B)	$(x \wedge \neg y) \Rightarrow (\neg x \Rightarrow y)$
(C)	$(\neg x \wedge y) \Rightarrow (\neg x \Rightarrow y)$
(D)	$(x \wedge \neg y) \Rightarrow (y \Rightarrow x)$

Q.30	Consider sorting the following array of integers in ascending order using an in-place Quicksort algorithm that uses the last element as the pivot. <table><tr><td>60</td><td>70</td><td>80</td><td>90</td><td>100</td></tr></table> The minimum number of swaps performed during this Quicksort is _____.	60	70	80	90	100
60	70	80	90	100		

Q.31

Consider the following two tables named Raider and Team in a relational database maintained by a Kabaddi league. The attribute *ID* in table Team references the primary key of the Raider table, *ID*.

Raider			
<i>ID</i>	<i>Name</i>	<i>Raids</i>	<i>RaidPoints</i>
1	Arjun	200	250
2	Ankush	190	219
3	Sunil	150	200
4	Reza	150	190
5	Pratham	175	220
6	Gopal	193	215

Team		
<i>City</i>	<i>ID</i>	<i>BidPoints</i>
Jaipur	2	200
Patna	3	195
Hyderabad	5	175
Jaipur	1	250
Patna	4	200
Jaipur	6	200

The SQL query described below is executed on this database:

```
SELECT *
FROM Raider, Team
WHERE Raider.ID=Team.ID AND City="Jaipur" AND
RaidPoints > 200;
```

The number of rows returned by this query is _____.

Q.32	The fundamental operations in a double-ended queue D are: <code>insertFirst(e)</code> – Insert a new element e at the beginning of D. <code>insertLast(e)</code> – Insert a new element e at the end of D. <code>removeFirst()</code> – Remove and return the first element of D. <code>removeLast()</code> – Remove and return the last element of D. In an empty double-ended queue, the following operations are performed: <code>insertFirst(10)</code> <code>insertLast(32)</code> <code>a ← removeFirst()</code> <code>insertLast(28)</code> <code>insertLast(17)</code> <code>a ← removeFirst()</code> <code>a ← removeLast()</code> The value of a is _____.
Q.33	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = \frac{1}{1+e^{-x}}$. The value of the derivative of f at x where $f(x) = 0.4$ is _____ (rounded off to two decimal places). Note: $\mathbb{R}$ denotes the set of real numbers.

Q.34	The sample average of 50 data points is 40. The updated sample average after including a new data point taking the value of 142 is _____.
------	---

Q.35	Consider the 3×3 matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$. The determinant of $(M^2 + 12M)$ is _____.
------	---

Q.36	A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently. What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?
(A)	2
(B)	4
(C)	6
(D)	8

Q.37	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. Note: $\mathbb{R}$ denotes the set of real numbers. $f(x) = \begin{cases} -x, & \text{if } x < -2 \\ ax^2 + bx + c, & \text{if } x \in [-2, 2] \\ x, & \text{if } x > 2 \end{cases}$ Which ONE of the following choices gives the values of a, b, c that make the function f continuous and differentiable?
(A)	$a = \frac{1}{4}, b = 0, c = 1$
(B)	$a = \frac{1}{2}, b = 0, c = 0$
(C)	$a = 0, b = 0, c = 0$
(D)	$a = 1, b = 1, c = -4$

Q.38	Consider the following Python code: <pre>def count(child_dict, i): if i not in child_dict.keys(): return 1 ans = 1 for j in child_dict[i]: ans += count(child_dict, j) return ans child_dict = dict() child_dict[0] = [1,2] child_dict[1] = [3,4,5] child_dict[2] = [6,7,8] print(count(child_dict, 0))</pre> Which ONE of the following is the output of this code?
(A)	6
(B)	1
(C)	8
(D)	9

Q.39	Consider the function computeS (X) whose pseudocode is given below: <pre> computeS (X) $S[1] \leftarrow 1$ for $i \leftarrow 2$ to $length(X)$ $S[i] \leftarrow 1$ if $X[i - 1] \leq X[i]$ $S[i] \leftarrow S[i] + S[i - 1]$ end if end for return S </pre> Which ONE of the following values is returned by the function computeS (X) for $X = [6, 3, 5, 4, 10]$?
(A)	[1, 1, 2, 3, 4]
(B)	[1, 1, 2, 3, 3]
(C)	[1, 1, 2, 1, 2]
(D)	[1, 1, 2, 1, 5]

Q.40	Let $F(n)$ denote the maximum number of comparisons made while searching for an entry in a sorted array of size n using binary search. Which ONE of the following options is TRUE?
(A)	$F(n) = F(\lfloor n/2 \rfloor) + 1$
(B)	$F(n) = F(\lfloor n/2 \rfloor) + F(\lceil n/2 \rceil)$
(C)	$F(n) = F(\lfloor n/2 \rfloor)$
(D)	$F(n) = F(n - 1) + 1$

Q.41	Consider the following Python function: <pre>def fun(D, s1, s2): if s1 < s2: D[s1], D[s2] = D[s2], D[s1] fun(D, s1+1, s2-1)</pre> What does this Python function <code>fun()</code> do? Select the ONE appropriate option below.
(A)	It finds the smallest element in D from index $s1$ to $s2$, both inclusive.
(B)	It performs a merge sort in-place on this list D between indices $s1$ and $s2$, both inclusive.
(C)	It reverses the list D between indices $s1$ and $s2$, both inclusive.
(D)	It swaps the elements in D at indices $s1$ and $s2$, and leaves the remaining elements unchanged.

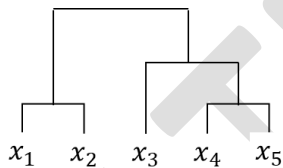
Q.42

Consider the table below, where the $(i, j)^{th}$ element of the table is the distance between points x_i and x_j . Single linkage clustering is performed on data points, x_1, x_2, x_3, x_4, x_5 .

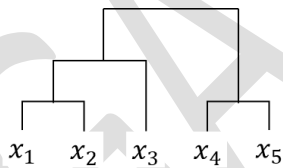
	x_1	x_2	x_3	x_4	x_5
x_1	0	1	4	3	6
x_2	1	0	3	5	3
x_3	4	3	0	2	5
x_4	3	5	2	0	1
x_5	6	3	5	1	0

Which **ONE** of the following is the correct representation of the clusters produced?

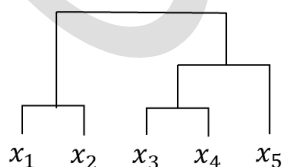
(A)



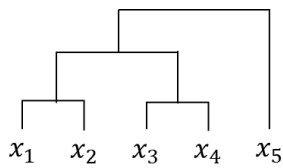
(B)



(C)



(D)



Q.43

Consider the two neural networks (NNs) shown in Figures 1 and 2, with $ReLU$ activation ($ReLU(z) = \max\{0, z\}, \forall z \in \mathbb{R}$). $\mathbb{R}$ denotes the set of real numbers. The connections and their corresponding weights are shown in the Figures. The biases at every neuron are set to 0. For what values of p, q, r in Figure 2 are the two NNs equivalent, when x_1, x_2, x_3 are positive?

Figure 1

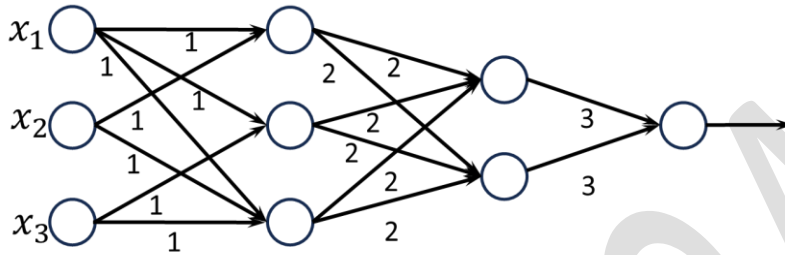
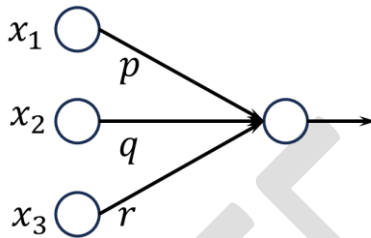


Figure 2



(A) $p = 36, q = 24, r = 24$

(B) $p = 24, q = 24, r = 36$

(C) $p = 18, q = 36, r = 24$

(D) $p = 36, q = 36, r = 36$

Q.44	Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered $n+1$ and $n+2$. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list. Which ONE of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?
(A)	BFS expands more states than DFS.
(B)	DFS expands more states than BFS.
(C)	Both BFS and DFS expand equal number of states.
(D)	Both BFS and DFS do not reach the goal state number 6.

Q.45	Consider the following sorting algorithms: (i) Bubble sort (ii) Insertion sort (iii) Selection sort Which ONE among the following choices of sorting algorithms sorts the numbers in the array [4, 3, 2, 1, 5] in increasing order after exactly two passes over the array?
(A)	(i) only
(B)	(iii) only
(C)	(i) and (iii) only
(D)	(ii) and (iii) only

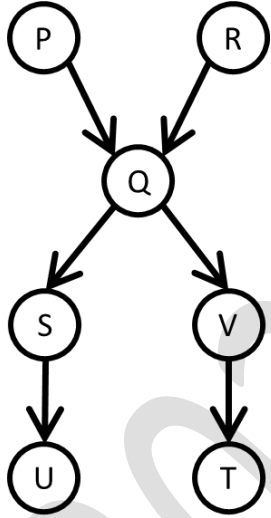
Q.46	Given the relational schema $R = (U, V, W, X, Y, Z)$ and the set of functional dependencies: $\{U \rightarrow V, U \rightarrow W, WX \rightarrow Y, WX \rightarrow Z, V \rightarrow X\}$ Which of the following functional dependencies can be derived from the above set?
(A)	$VW \rightarrow YZ$
(B)	$WX \rightarrow YZ$
(C)	$VW \rightarrow U$
(D)	$VW \rightarrow Y$

Q.47	Select all choices that are subspaces of $\mathbb{R}^3$. Note: $\mathbb{R}$ denotes the set of real numbers.
(A)	$\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
(B)	$\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
(C)	$\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0 \right\}$
(D)	$\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

Q.48	Which of the following statements is/are TRUE ? Note: $\mathbb{R}$ denotes the set of real numbers.
(A)	There exist $\mathbf{M} \in \mathbb{R}^{3 \times 3}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{M}\mathbf{x} = \mathbf{p}$ has a unique solution and $\mathbf{M}\mathbf{x} = \mathbf{q}$ has infinite solutions.
(B)	There exist $\mathbf{M} \in \mathbb{R}^{3 \times 3}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{M}\mathbf{x} = \mathbf{p}$ has no solutions and $\mathbf{M}\mathbf{x} = \mathbf{q}$ has infinite solutions.
(C)	There exist $\mathbf{M} \in \mathbb{R}^{2 \times 3}$, $\mathbf{p} \in \mathbb{R}^2$, and $\mathbf{q} \in \mathbb{R}^2$ such that $\mathbf{M}\mathbf{x} = \mathbf{p}$ has a unique solution and $\mathbf{M}\mathbf{x} = \mathbf{q}$ has infinite solutions.
(D)	There exist $\mathbf{M} \in \mathbb{R}^{3 \times 2}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{M}\mathbf{x} = \mathbf{p}$ has a unique solution and $\mathbf{M}\mathbf{x} = \mathbf{q}$ has no solutions.

Q.49	Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$ and $\mathbf{M} \in \mathbb{R}^{3 \times 3}$ be the matrix corresponding to the projection on to the subspace U. Which of the following statements is/are TRUE?
(A)	If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of $\mathbf{M}$ is a 1-dimensional subspace.
(B)	If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of $\mathbf{M}$ is a 1-dimensional subspace.
(C)	$\mathbf{M}^2 = \mathbf{M}$
(D)	$\mathbf{M}^3 = \mathbf{M}$

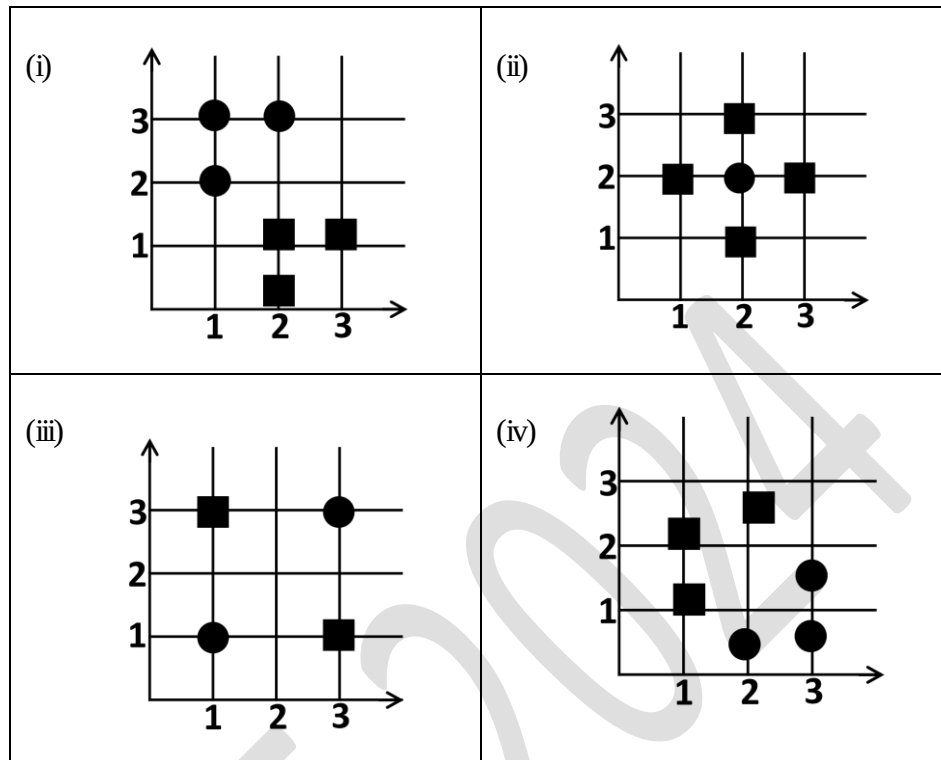
Q.50	Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ where $\mathbb{R}$ is the set of all real numbers. $f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$ Which of the following statements is/are TRUE?
(A)	$x = 0$ is a local maximum of f
(B)	$x = 3$ is a local minimum of f
(C)	$x = -1$ is a local maximum of f
(D)	$x = 0$ is a local minimum of f

Q.51	Consider the directed acyclic graph (DAG) below:  <pre> graph TD P((P)) --> Q((Q)) R((R)) --> Q((Q)) Q((Q)) --> S((S)) Q((Q)) --> V((V)) S((S)) --> U((U)) V((V)) --> T((T)) </pre> Which of the following is/are valid vertex orderings that can be obtained from a topological sort of the DAG?
(A)	P Q R S T U V
(B)	P R Q V S U T
(C)	P Q R S V U T
(D)	P R Q S V T U

Q.52	Let H, I, L, and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree. Which of the following statements is/are always TRUE?
(A)	$L \leq I + 1$
(B)	$H + 1 \leq N \leq 2^{H+1} - 1$
(C)	$H \leq I \leq 2^H - 1$
(D)	$H \leq L \leq 2^{H-1}$

Q.53

Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and squares.



Which of the following is/are **TRUE**?

(A) (i) is linearly separable.

(B) (ii) is linearly separable.

(C) (iii) is linearly separable.

(D) (iv) is linearly separable.

Q.54	Let $game(ball, rugby)$ be true if the ball is used in rugby and false otherwise. Let $shape(ball, round)$ be true if the ball is round and false otherwise. Consider the following logical sentences: $s1: \forall ball \neg game(ball, rugby) \Rightarrow shape(ball, round)$ $s2: \forall ball \neg shape(ball, round) \Rightarrow game(ball, rugby)$ $s3: \forall ball game(ball, rugby) \Rightarrow \neg shape(ball, round)$ $s4: \forall ball shape(ball, round) \Rightarrow \neg game(ball, rugby)$ Which of the following choices is/are logical representations of the assertion, “All balls are round except balls used in rugby”?
(A)	$s1 \wedge s3$
(B)	$s1 \wedge s2$
(C)	$s2 \wedge s3$
(D)	$s3 \wedge s4$

Q.55	An OTT company is maintaining a large disk-based relational database of different movies with the following schema: <pre> Movie(<u>ID</u>, CustomerRating) Genre(<u>ID</u>, Name) Movie_Genre(<u>MovieID</u>, <u>GenreID</u>) </pre> Consider the following SQL query on the relation database above: <pre> SELECT * FROM Movie, Genre, Movie_Genre WHERE Movie.CustomerRating > 3.4 AND Genre.Name = "Comedy" AND Movie_Genre.MovieID = Movie.ID AND Movie_Genre.GenreID = Genre.ID; </pre> This SQL query can be sped up using which of the following indexing options?
A	B ⁺ tree on all the attributes.
B	Hash index on Genre.Name and B ⁺ tree on the remaining attributes.
C	Hash index on Movie.CustomerRating and B ⁺ tree on the remaining attributes.
D	Hash index on all the attributes.

Q.56	Let X be a random variable uniformly distributed in the interval $[1, 3]$ and Y be a random variable uniformly distributed in the interval $[2, 4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).
------	---

Q.57	Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by: $f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$ If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ indicate the expectation and variance of X, respectively, the value of λ is _____ (rounded off to one decimal place).
Q.58	Consider two events T and S. Let $\bar{T}$ denote the complement of the event T. The probability associated with different events are given as follows: $P(\bar{T}) = 0.6, \quad P(S T) = 0.3, \quad P(S \bar{T}) = 0.6$ Then, $P(T S)$ is _____ (rounded off to two decimal places).
Q.59	Consider a joint probability density function of two random variables X and Y $f_{X,Y}(x, y) = \begin{cases} 2xy, & 0 < x < 2, \quad 0 < y < x \\ 0, & \text{otherwise} \end{cases}$ Then, $E[Y X = 1.5]$ is _____.
Q.60	Evaluate the following limit: $\lim_{x \rightarrow 0} \frac{\ln((x^2 + 1) \cos x)}{x^2} = \text{_____}.$

Q.61

Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix

$\mathbf{M} = \mathbf{u}\mathbf{u}^T$ (where $\mathbf{u}^T$ is the transpose of $\mathbf{u}$). The value of $\sum_{i=1}^5 \sigma_i$ is _____.

Q.62

Details of ten international cricket games between two teams “Green” and “Blue” are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute *Pitch* can take one of two values: spin-friendly (represented as *S*) or pace-friendly (represented as *F*). The attribute *Format* can take one of two values: one-day match (represented as *O*) or test match (represented as *T*).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

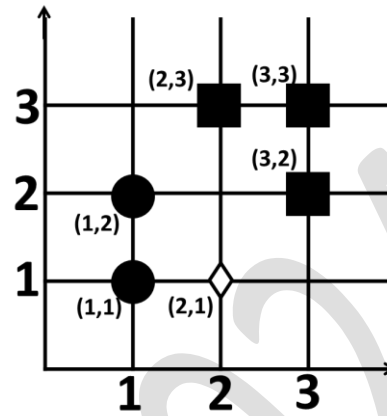
To develop such a model, the computed $\text{InformationGain}(\mathbf{C}, \text{Pitch})$ with respect to the Target is _____ (rounded off to **two** decimal places).

Table C

Match Number	Pitch	Format	Winner (Target)
1	<i>S</i>	<i>T</i>	Green
2	<i>S</i>	<i>T</i>	Blue
3	<i>F</i>	<i>O</i>	Blue
4	<i>S</i>	<i>O</i>	Blue
5	<i>F</i>	<i>T</i>	Green
6	<i>F</i>	<i>O</i>	Blue
7	<i>S</i>	<i>O</i>	Green
8	<i>F</i>	<i>T</i>	Blue
9	<i>F</i>	<i>O</i>	Blue
10	<i>S</i>	<i>O</i>	Green

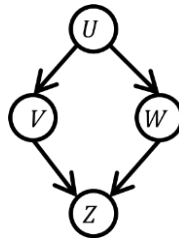
Q.63

Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k -nearest neighbor algorithm for which the diamond ($\diamond$) shaped data point is assigned the label square is _____.



Q.64

Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \cdot)$	$P(V = 1 \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \cdot)$	$P(W = 1 \cdot)$
$U = 0$	1	0
$U = 1$	0	1

		$P(Z = 0 \cdot)$	$P(Z = 1 \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) = \underline{\hspace{2cm}}$ (rounded off to **three** decimal places).

Q.65

Two fair coins are tossed independently. X is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. Y is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise.

The value of the covariance of X and Y is $\underline{\hspace{2cm}}$ (rounded off to **three** decimal places).